

RadiFlux: A user-friendly graphical interface to model physical processes in ice-covered lakes

Undergraduate: Ari Bloch

Mentor: Dr. Benjamin Smith

Abstract

Accurate modeling of the water temperature in freshwater ice-covered lakes is critical for predicting the movement of phytoplankton and nutrients driven by solar-powered water motion. As light is allowed to propagate through the ice, it is absorbed and heats the near-ice water increasing both the temperature and density of cold water to form a mixing layer at the ice cover. The goal is to create a user-friendly webpage to allow users to change external parameters such as solar intensity, along with lake characteristics such as depth and ice composition, to predict the onset of this mixing. Utilizing advanced mathematics, the webpage will be built using Python, with an easy-to-use graphical interface to observe the evolution of the water temperature and to predict the onset of solar-powered mixing from user-input data.

1. Introduction

In winter, many freshwater lakes in the northern hemisphere (e.g. Great Lakes region) freeze over.

However, they are not dormant, as they become hot spots for biological (e.g. phytoplankton) and physical processes (e.g. water motion) (Jansen et al, 2021). These often influence one another, for example

Radiatively Driven Convection (RDC) can result from sunlight penetrating through the ice, heating near-ice and lead mixing as increasing water temperatures below 4 °C increases its density to produce a gravitationally unstable water column. This mixing creates a mechanism that transports phytoplankton to the ice and promote growth (Smith et al, 2025). A recent study by Smith et al (2025) found that the onset of RDC could be predicted by analyzing the evolution of the temperature profile. Using high resolution computer simulations, they provide an equation that reasonably predicts the onset of RDC which could be used to gather information about the ecosystem beneath. Current tools require expertise, making it difficult for users, ranging from Limnologists to the general public, to approach. The goal of this project is to predict the onset of RDC from user-input data with an easy-to-use and understandable graphical user interface (GUI).

Recent interest, such as RDC, has stemmed from a lack of understanding of the winter period that was historically overlooked due to adverse conditions (Jansen et al 2021). The first observations of RDC include Farmer (1975) and Mironov (2002), who, by explicitly including solar radiation in mathematical models, were among the first to make the connection between RDC and the under-ice water temperature.

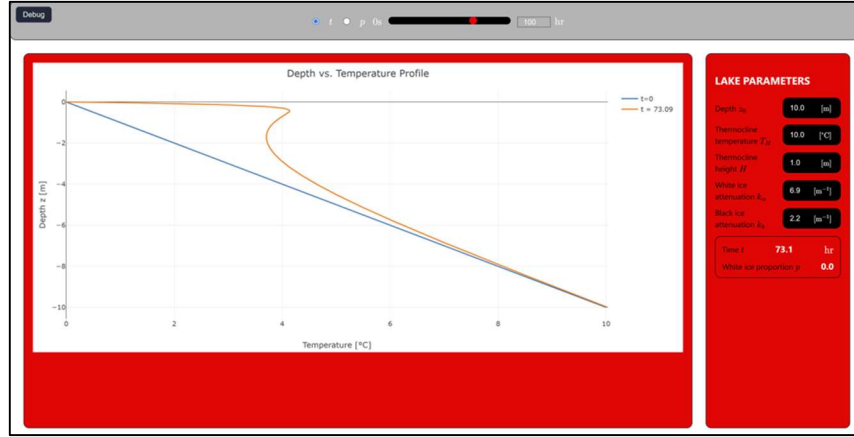


Figure 1: Our prototype that displays the under-ice water temperature heated by solar radiation. Users can add custom parameters (right) and vary the sliders (top) to visualize how the temperature evolves with time (orange line) with respect to an initial state (blue line).

One of the biggest tools currently used to study RDC in lakes are computational simulations or computational fluid dynamics (CFD) (e.g. see Allum et al, 2024). However, it requires specialist knowledge including fluid dynamics and advanced mathematics. Additionally, it requires a significant amount of computational resources meaning that CFD can be costly and less accessible. Another approach is to conduct field experiments to collect physical samples and measurements (e.g. see Bouffard et al, 2019). Although crucial, it can take a long time to collect and analyze the data as well as being expensive to obtain the necessary equipment. Overall, these methodologies are incredibly useful, however they are expensive and difficult to implement without specialized knowledge.

Our solution is significantly cheaper and more accessible. Over the last academic year, we have developed a prototype (figure 1), utilizing the models outlined in Smith (2025), of a webpage that displays a user-friendly GUI of the under-ice water temperature. The user can input different lake parameters such as the depth of the lake (z_0), the depth (H) of the thermocline i.e. the region where the near-ice water temperature linearly increases from 0 °C to a temperature T_H , and how well light permeates through cloudy white (k_w) and transparent black ice (k_b) which lets less or more light through respectively. As seen in figure 1, this generates the temperature profile at an initial time $t = 0$ hrs. The user can input a maximum time and can drag the slider to see how the temperature will change over time. They can also change the ice composition (ratio of white ice to black ice) p via a slider, which will vary

the amount of light available to heat the water. Although there is already a large variety of software that models water temperature to varying degrees of complexity, for example, the LakeModelR package in R (Ladwig, 2022). However, these often require pre-existing knowledge, such as R, and provide no obvious graphical ability, whereas our prototype changes dynamically as the user changes input.

2. Research Questions and Objectives

As discussed above, there is a lack of user-friendly tools to observe and analyze physical processes underneath winter ice cover. In response, we will tackle this by extending the capabilities of our prototype with the following objectives:

I. Incorporate user input to upload a custom initial temperature profile. Currently, our prototype assumes a fixed profile which the user can customize (figure 1). However, by including an option for the user to upload a custom initial profile (e.g. via field work), then they can see how their profile evolves in time without the need for high frequency measurements. Additionally, this would allow users to obtain predictions for the onset of RDC (see objective II) and aid users (see objective III) to interpret the results.

II. Predict the onset of RDC for a given temperature profile. Smith et al (2025) provided an equation to determine the onset of RDC based on the initial temperature profile and environmental inputs (e.g. water clarity and under-ice light intensity). By incorporating this functionality, it will aid users to understand how long is needed until mixing can begin.

III. Further update the GUI to aid visualization of under-ice physical processes. To incorporate objectives I and II, further updates to the GUI are necessary, specifically, adding a simple method to upload a custom initial profile (objective I). Additionally, we would also like to add different optional markers (via toggle) that would show the onset of RDC for a more tailored experience.

3. Methodology

To achieve our objectives, we will use our prototype (figure 1) to create an interactive webpage using Python, JavaScript, HTML, and CSS. In the prototype, we have already implemented the models outlined

in Smith et al (2025). However, our goal is to incorporate field data to make accurate predictions for the onset of RDC. To achieve this, we will add functionality for the user to upload a custom initial profile (objective I) via a file upload (e.g. csv file). An initial profile will be created from an interpolant of the uploaded data that can then evolve with time. We will then use data already collected from Lake Mendota (Madison, WI) by Dr. Smith to test the functionality and accuracy of the program.

To predict the onset of RDC (objective II), we will use the predictor of Smith et al (2025). This depends on the initial temperature profile and environmental factors, such as the under-ice light intensity, via a calculable forcing parameter that dictates whether solar forcing can overcome diffusive processes to produce a gravitationally unstable water column. We will also use the field data to test this predictor. Finally, to aid visualization (objective III) we will implement a simple file upload to allow users to upload different data types and to automatically produce an initial profile. There will also be different toggles to allow users to understand their data.

4. Timeline

We propose to complete this project over a period of 8 weeks over the summer of 2026 (figure 2). The first two weeks will be spent researching how to turn our prototype (figure 1) into a functional webpage and how to create an interpolant based on user-uploaded data. This will then transition into experimenting with different methods to upload user data and to interpolate the data for use in our original prototype. The majority of the summer would then be spent implementing these additions. Once the webpage is launched, we will test the output to field data collected from this past winter to verify the models we used. Finally, over the last three weeks, we will complete the final report to disseminate our findings.

5. Conclusion

In summary, interest in freshwater ice-covered lakes is growing, yet current tools are not accessible for general users. In response, over the course of 8 weeks in the summer of 2026, we propose to create an interactive webpage with an easy-to-use GUI to predict the onset of RDC via the under-ice water temperature from user-input data. To achieve this, we will upgrade our working prototype to include a

OBJECTIVES	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
RESEARCH								
Interpolation								
Web page hosting								
File upload								
IMPLEMENT								
Prototype user-input methods								
GUI								
RDC predictions								
Launch website								
Fit user-data								
TEST								
Compare to field measurements								
REPORT								
Write Report								

Figure 2: Gantt chart showing the timeline of the project over 8 weeks in the summer of 2026.

functionality for users to upload their own data (e.g. field measurements) to observe the evolution of the water temperature under sustained solar heating. To incorporate complex physical processes, we will also implement predictions of the onset of RDC. Finally, by upgrading the GUI, we will have an easy-to-use GUI for users to interact with and further understand ice-covered freshwater lakes.

References

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